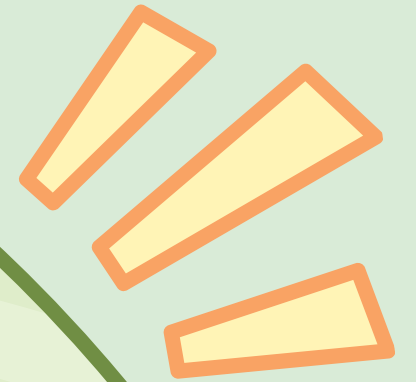




Soil Layers – Make a Core Sample



What We Will Learn Today

Today we will discover exciting things about soil

- Name the 3 layers of soil
- Make our own core sample
- Discover what lives in soil
- Get ready to become soil scientists!





What is Under Our Feet?

Have you ever wondered what is below the ground where we play? Let's find out together!



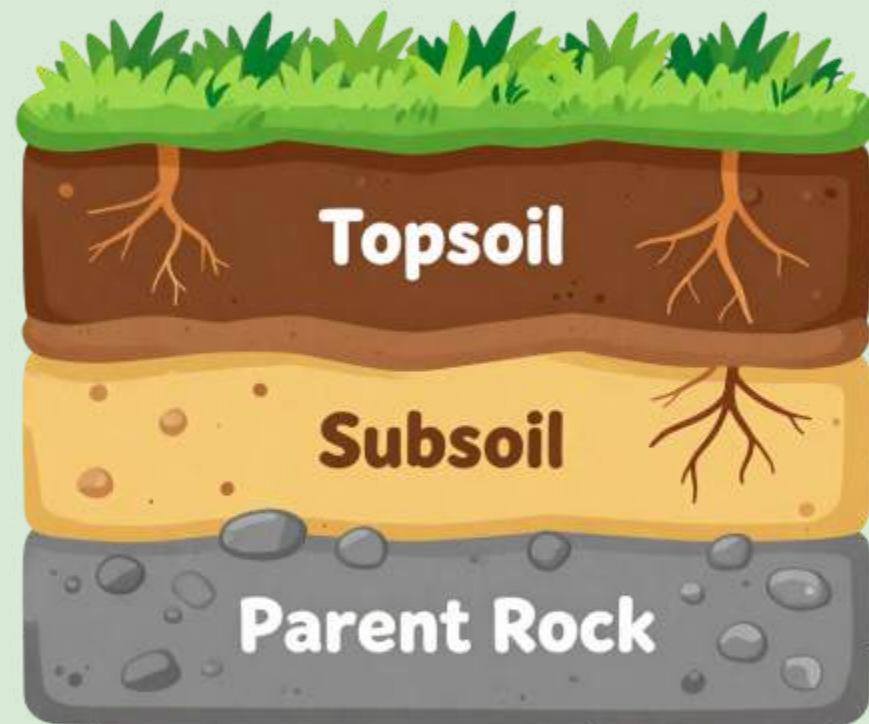


Soil is Like a Layered Barfi!

Just like pastry has different colored layers, the ground also has layers! Each layer is different and special.
Look at the picture — can you see how the sweet and the soil both have layers stacked on top of each other?



The 3 Layers of Soil



- TOP LAYER (Topsoil) — Dark and soft, this is where plants grow and tiny creatures live! It feels crumbly like chocolate cake.
- MIDDLE LAYER (Subsoil) — Light brown or yellow colored, sticky like clay. Big tree roots reach down here to drink water!
- ROCK LAYER (Parent Rock) — Hard stones and pebbles at the very bottom. This is the strong foundation that holds everything up!





Top Layer – Topsoil



- Dark brown or black color — like the rich soil in your school garden!
- Soft and crumbly — easy to dig with your fingers
- Where plants grow and worms live! The roots of tulsi, mango trees, and vegetables all start here. Earthworms make their homes in topsoil too!





Middle Layer — Subsoil

- Lighter color — yellow or light brown
- More clay, sticky and dense
- Big tree roots reach here!
- Indian example: Lighter soil visible in road cutting in Kerala or Karnataka hills





Rock Layer — Parent Rock

- Grey or tan color — looks like sand and stones mixed together
- Hard pebbles and stones that feel rough
- The foundation of everything above — like the bottom of a building!

Indian Example: You can see rocky pieces at construction sites or on the famous Deccan Plateau where big rocks sit under the soil!





What is a Core Sample?

A core sample is like pushing a straw into a layered sandwich! Scientists use it to see all the layers without digging a big hole. Instead of digging up the whole ground, we can take a small sample and see everything inside!





Let's Make Our Core Sample!



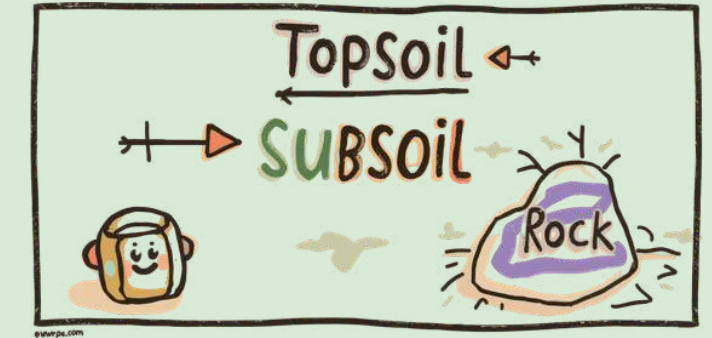
Step 1: Add gravel and pebbles at the **BOTTOM** of your bottle. This is the Rock Layer — hard and strong!

Step 2: Add light sand or clay in the **MIDDLE**. This is the Subsoil — lighter and sticky!





Let's Make Our Core Sample! (Part 2)



Step 3: Add Dark Soil on TOP
This is the TOPSOIL layer!
It's dark brown and soft.
Put it right on top of your middle layer. This is where plants grow!

Step 4: Add a Tiny Decoration
Place a small leaf or a toy worm on top! This shows what lives in topsoil.
Earthworms love it here!

Step 5: Label Each Layer!
Use stickers or paper to label: TOPSOIL at top, SUBSOIL in middle, ROCK at bottom. Your core sample is complete!



What Lives in Topsoil?



- Earthworms — Farmer's friends! They dig tunnels and make soil healthy for plants to grow.
- Ants — Tiny tunnel builders! They carry food underground and help mix the soil layers.
- Plant roots — Mango, neem, and tulsi roots drink water and hold tight to the topsoil!





Let's Remember Together!

What's on TOP?

→ TOPSOIL! Dark and soft!

What's in the MIDDLE?

→ SUBSOIL! Clay and light!

What's at the BOTTOM?

→ ROCK! Hard and tight!

What do we call our science tube?

→ A CORE SAMPLE! Wah!





Take This Home!

Tonight, ask someone at home:
"When we plant tulsi or
tomatoes, which layer do the
seeds go in?"
The answer is... TOPSOIL!
Can you explain why?





**Thank You for Learning
About Soil Layers!**

